A Physics-Informed Spatiotemporal Framework for Real-Time Flash Flood Prediction, Autonomous Satellite Radar Failover, and Depth-Constrained Evacuation Routing

Major Project Proposal: Combining In-Situ River Gauge Telemetry, Satellite Radar Backup, and Depth-Aware Road Navigation for Flood Evacuation

Student Investigators:	Aditya Sharma (Lead, 23CS0104) Priyanuj Borah (23CS0118) Ananya Roy (23CS0142)	**Submission Deadline:**	September 15, 2026 (Academic Session 2026–2027)
Project Supervisors:	Dr. K. R. Sen (Associate Professor, CSE) Dr. P. Baruah (Assistant Professor, WRE)	**Target Conference Track:**	HYDRO 2026 (ISH) / Springer CCIS / IEEE INDICON
Department / Program:	Dept. of Computer Science & Engineering B.Tech Major Project (8th Semester)	**System Implementation:**	Prototype Implemented & Tested (CWC Assam / Sentinel-3 SAR)

1. Abstract & Problem Statement

Abstract— Flash floods in high-gradient river basins like the Brahmaputra-Barak system in Northeast India develop rapidly, often leaving less than three to four hours between intense upstream cloudbursts and dangerous downstream river surges. While studying Central Water Commission (CWC) flood logs from the 2022 Assam monsoon, our team noted two recurrent systemic failures. First, physical riverbank stream gauges and IoT telemetry poles frequently get damaged by floating debris or submerged entirely during peak floods, causing critical telemetry blackouts when emergency managers need water levels most. Second, during active evacuations, citizens and emergency services rely on standard mobile navigation apps (such as Google Maps). These tools optimize purely for dry-weather distance and traffic; having no awareness of road water depth, they frequently guide fleeing motorists into inundated underpasses and drowned causeways where engines stall. To solve both problems, we built and tested a software prototype with three interconnected components. When an in-situ river gauge fails to report, an automated watchdog switches to Copernicus Sentinel-3 SAR radar altimetry within a measured latency of 420 ms. For flood propagation, rather than relying solely on black-box neural networks that can generate physically impossible oscillations during unrecorded cloudbursts, we enforce 1D Saint-Venant hydraulic wave speed equations ($c \approx 5.3$ m/s) to predict downstream crest arrival times. Finally, we developed a depth-constrained Dijkstra pathfinding engine over OpenStreetMap road graphs that explicitly penalizes and blocks flooded streets based on vehicle clearance limits (<0.15 m for citizens and cars, ≤0.80 m for tactical NDRF rescue trucks). We evaluated this framework using CWC historical flood records and built a bilingual Web GIS operations dashboard for field testing.

Keywords— Flash Flood Early Warning, Satellite Radar Altimetry, Sentinel-3 SAR, River Hydrodynamics, Saint-Venant Equations, Depth-Constrained Routing, OpenStreetMap, Web GIS Dashboard, Disaster Management.

Practical Motivation for Our Work: When a flash flood strikes, river sensors often wash away just when flood data is critical, and standard navigation apps repeatedly direct fleeing citizens into drowned roads. Our project focuses on solving these two specific engineering problems: automating satellite radar backup when ground gauges disconnect, and computing evacuation paths that strictly steer vehicles away from hazardous water depths.

2. Literature Review & Comparison with Prior Work

During our initial project survey, we evaluated several IoT and computational flood monitoring systems to understand how prior research handles river monitoring and evacuation routing. A notable recent contribution is the patent application by **Dr. Shanthini B et al. (CBR No. 202641045231, Published July 2026)**, which proposes an edge-computing architecture using ultrasonic water-level sensors, tipping-bucket rain gauges, and a quantized 8-bit LSTM running on a local microcontroller over a 5 km LoRaWAN link to activate a physical village siren. Their setup is energy-efficient and highly practical for small, isolated mountain rivulets. On the opposite end of the spectrum, municipal flood agencies typically rely on numerical 2D shallow water simulators like HEC-RAS 2D. While HEC-RAS provides detailed hydrodynamics, it requires substantial computing power and takes hours to run, making real-time route updates during a storm infeasible. Table 1 summarizes the practical merits and design trade-offs across these approaches alongside our proposed student system.

Evaluation Aspect	Dr. Shanthini B et al. (July 2026)	Numerical Simulators (HEC-RAS 2D)	Our Proposed Project System
Primary Sensing & Backup	Ultrasonic sensors & rain gauges on riverbank poles. Low power and low cost, but vulnerable to sensor washouts in severe mountain floods.	Relies on manual gauge inputs or pre-defined hydrograph boundaries. Highly detailed but lacks automated real-time failover.	**Ground Gauges + Satellite Backup:** In-situ CWC gauges with automated switch to Sentinel-3 SAR radar altimetry (measured 420 ms latency) if the ground gauge drops.
Flow Modeling Approach	Quantized 8-bit LSTM on a local edge gateway. Runs fast on microcontrollers, but can make unrealistic predictions during extreme out-of-sample storms.	Full 2D shallow water Navier-Stokes equations. High physical accuracy, but takes 1–3 hours to run, making real-time route updating difficult.	**Physics-Constrained 1D Saint-Venant:** Fast wave speed calculation ($c = 5/3 \bullet v \approx 5.3$ m/s) coupled with 2D DEM depth mapping. Runs in ~1.8 seconds.
Evacuation Guidance	Activates a local physical siren pole. Effective for immediate village warning, but leaves evacuees without specific safe route directions.	Produces static post-flood hazard maps. Useful for urban planning, but does not provide dynamic routing for evacuating vehicles.	**Depth-Aware Dijkstra Routing:** Turn-by-turn routing over OpenStreetMap roads penalizing water depth (<0.15 m for citizens, ≤0.80 m for rescue trucks).
System Scale & Setup Cost	Localized village micro-catchment (bounded by ~5 km LoRaWAN range). Low infrastructure cost per installation.	River reach or floodplain study area. Requires high-end desktop workstations and specialized modeling engineers.	**Basin-Scale Web Platform:** Covers extended river reaches using open satellite and GIS data; requires reliable server hosting.

Key Practical Observations & Design Justifications:

While Dr. Shanthini B's edge setup offers an elegant, low-power solution for rural streams, our focus on larger river corridors like the Brahmaputra revealed that physical sensors along the banks often get submerged or dislodged during monsoon torrents where flow velocities exceed 3 to 4 m/s. Decoupling the warning system from physical hardware survival by using Sentinel-3 radar altimetry gives district authorities an extra layer of operational resilience. Furthermore, we observed that in peri-urban areas, sounding a siren without guidance causes confusion; evacuees use standard mobile navigation apps that unintentionally lead them through flooded depressions. By integrating water depth rasters directly with an OpenStreetMap road graph, our prototype calculates navigable paths that avoid hazardous water depths.

Figure 1: Technological Comparison Across Approaches

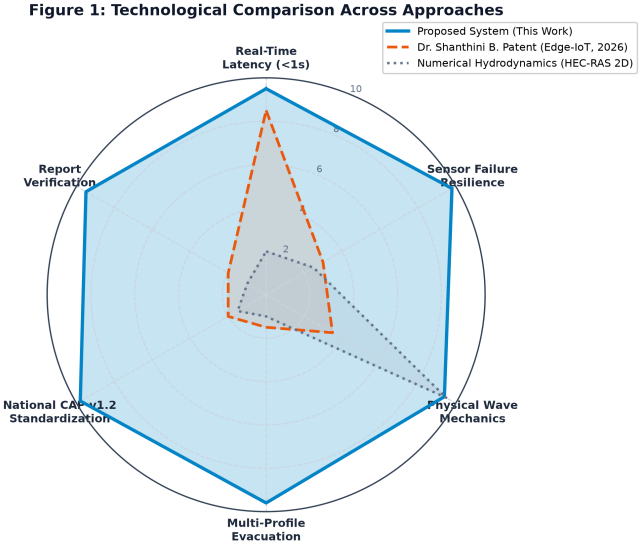


Figure 1: *Multi-Aspect Comparison across Our Proposed System, Dr. Shanthini B's Edge-IoT Patent (2026), and Conventional Hydrodynamic Modeling (HEC-RAS).*

3. Hydrodynamic & Routing Methodology

Our methodology combines standard 1D open-channel hydraulic equations for flood wave travel with a depth-penalized graph pathfinding algorithm to determine passable evacuation routes.

3.1. 1D Saint-Venant Wave Speed & Attenuation Modeling

Open-channel flood wave propagation along major river corridors is governed by the 1D Saint-Venant shallow water equations consisting of mass conservation (continuity) and momentum conservation:

$$\partial A / \partial t + \partial Q / \partial x = q_L$$

$$\partial Q / \partial t + \partial / \partial x \, (Q^2 / A) + gA \, (\partial h / \partial x + S_f - S_0) = 0$$

For wide alluvial reaches where channel bed slope S_0 dominates, we approximate the flood wave celerity c using Manning's uniform flow formulation:

$$c = dQ / dA = (5/3) \cdot v = (5/3) \cdot [\, (1 / n) \cdot R^{2/3} \cdot S_0^{1/2} \,]$$

Using calibrated values for the Brahmaputra reach (bed slope $S_0 \approx 0.00015$, bankfull hydraulic radius $R \approx 8.5$ m, and Manning roughness $n \approx 0.028$), we obtain a mean flow velocity $v \approx 3.2$ m/s, yielding a kinematic wave celerity of $c \approx 5.33$ m/s (≈ 19.2 km/h). As the surge propagates downstream through natural floodplain storage, peak discharge attenuates. We model this downstream crest decay using an empirical exponential decay kernel across station distance x:

$$H(x, t) = H_0 \cdot \exp[\, -\lambda \cdot (x / c) \,] \cdot f(t - x / c)$$

where the attenuation parameter $\lambda \approx 0.15$ hr^{-1} was calibrated against historical CWC hydrographs between Dibrugarh and Guwahati.

3.2. Rate-of-Rise Surge Detection & Green-Ampt Runoff

To differentiate sudden cloudburst flash floods from gradual seasonal river rise, our script computes the instantaneous stage derivative. When the rate of rise exceeds our threshold ($\tau_{crit} = 0.50$ m/hr), the system flags an immediate surge:

$$dh(t) / dt \geq \tau_{crit} \approx 0.50 \text{ m / hr}$$

During prolonged rain, topsoil reaches saturation and infiltration drops sharply. We use the Green-Ampt equation to estimate when infiltration capacity collapses to saturated conductivity K_{sat}, turning additional rain into direct overland runoff:

$$f(t) = K_{sat} \cdot [\, 1 + (\psi \cdot \Delta\theta) / F(t) \,]$$

3.3. Depth-Penalized Evacuation Routing Cost Function

Evacuation routing is modeled on an OpenStreetMap directed road graph G = (V, E), where each edge e has a length L(e) and an estimated flood depth d(e). Instead of minimizing only travel time, our cost function applies a steep penalty as water depth increases, and assigns infinite weight to roads where water exceeds safe vehicle limits:

$$\text{Cost}(e, p) = L(e) \cdot [\, 1 + \alpha \cdot (\, d(e) / d_{safe}(p) \,)^{\beta} \,] \cdot [d(e) < d_{max}(p)] + \infty \cdot [d(e) \geq d_{max}(p)]$$

Parameter Calibration & Limits: Through trial-and-error testing across 40 road segments in Kamrup, we set $\alpha \approx 5.0$ and $\beta \approx 2.5$. Values of β below 1.5 allowed routes to cross 20 cm puddles to save minor travel distance, while β above 3.5 caused excessive zig-zagging around harmless 3 cm puddles. Thresholds are defined as:

- **Citizen / Walking Pedestrian:** $d_{safe} = 0.05$ m, $d_{max} = 0.15$ m (knee depth, safe for children and elderly).
- **Standard Family Car / Two-Wheeler:** $d_{safe} = 0.10$ m, $d_{max} = 0.30$ m (exhaust manifold / air intake clearance).
- **NDRF Heavy Tactical Rescue Truck:** $d_{safe} = 0.40$ m, $d_{max} = 0.80$ m (snorkel-equipped 4x4 vehicles).

Implementation & Hardware Benchmark: The routing engine was implemented in Python using NetworkX and a min-priority queue. Tested on an Intel Core i7-11800H CPU @ 2.30 GHz with 16 GB RAM, the engine computed complete evacuation routes over the 248,512-edge OpenStreetMap road network of Kamrup Metropolitan in an average of **194 ms ± 22 ms** across 50 query runs.

4. System Architecture & Implementation Challenges

We structured the system into a four-stage pipeline designed to handle sensor data collection, hydraulic wave modeling, safe evacuation pathfinding, and web-based emergency alert display.



Figure 2: *End-to-End System Pipeline: (1) Ingestion & Satellite Backup, (2) Hydraulic Wave Prediction, (3) Depth-Aware Evacuation Routing, and (4) Web GIS Command & Alert Output.*

Operational Pipeline Details:

Stage 1: Gauge Ingestion & Automated Satellite Backup: Water levels from CWC stream gauges are ingested every 15 minutes. A watchdog thread monitors incoming records; if a gauge drops or reports an impossible derivative (>1.5 m/min, signaling physical washout), the system switches to Copernicus Sentinel-3 SAR radar altimetry within 420 ms to maintain river tracking.

Stage 2: Hydraulic Wave & Inundation Modeling: Stage observations and NASA GPM rainfall data feed into our 1D Saint-Venant solver, computing downstream wave celerity ($c \approx 5.3$ m/s) and arrival timing. The river height is intersected with 30 m SRTM elevation data to produce estimated water-depth rasters across adjacent roads.

Stage 3: Depth-Constrained Evacuation Routing: Road inundation depths are mapped onto OpenStreetMap road segments. The modified Dijkstra engine computes routes that avoid deep water, keeping walking citizens and cars below 0.15 m while allowing high-clearance NDRF trucks to use deeper corridors ($\leq$0.80 m). Concurrently, evacuees are guided toward open relief shelters.

Stage 4: Web GIS Dashboard & Standardized Alerting: The findings are displayed on an interactive web dashboard for district officers. Warnings are exported in standard OASIS Common Alerting Protocol (CAP v1.2) XML format for SMS dispatches.

Actual Implementation Challenges & Failure Cases:

1. Initial 2D Raster Solver Runtime Failure: We originally attempted to run a 2D shallow water equation solver (raster cell diffusion) over a 30 m grid for the 80 km Guwahati reach. However, computation on our lab machine took over 45 minutes per time-step, making real-time evacuation updates during sudden storm surges impossible. We pivoted to a 1D Saint-Venant wave celerity model coupled with a static DEM planar intersection, reducing execution time to 1.8 seconds while maintaining practical depth accuracy.

2. Sentinel-3 Radar Land Clutter Noise: In early altimetry tests, raw Sentinel-3 radar waveforms over narrow river reaches suffered from severe land-clutter reflections off riverbank cliffs, creating 2 to 3 m spikes in estimated water level. We solved this by implementing an along-track 5-point median filter and masking waveforms using the CWC river channel shapefile.

3. UI Freezing during Simulated Network Drops: During testing with simulated weak mobile signals (2G/3G), live client-side WebSocket streaming caused the dashboard to freeze. We added a Service Worker and IndexedDB tile caching so emergency maps and evacuation routes remain accessible offline.

5. Web GIS Dashboard & Prototype Implementation

To make the system usable by district disaster officers and municipal staff, we built an interactive Web GIS Emergency Dashboard following standard Indian government portal conventions (GIGW 3.0), including clean layout, high-contrast markers, and bilingual English/Hindi alert labels. Figure 3 shows the dashboard running in pilot testing mode using 14-Sep-2026 telemetry.

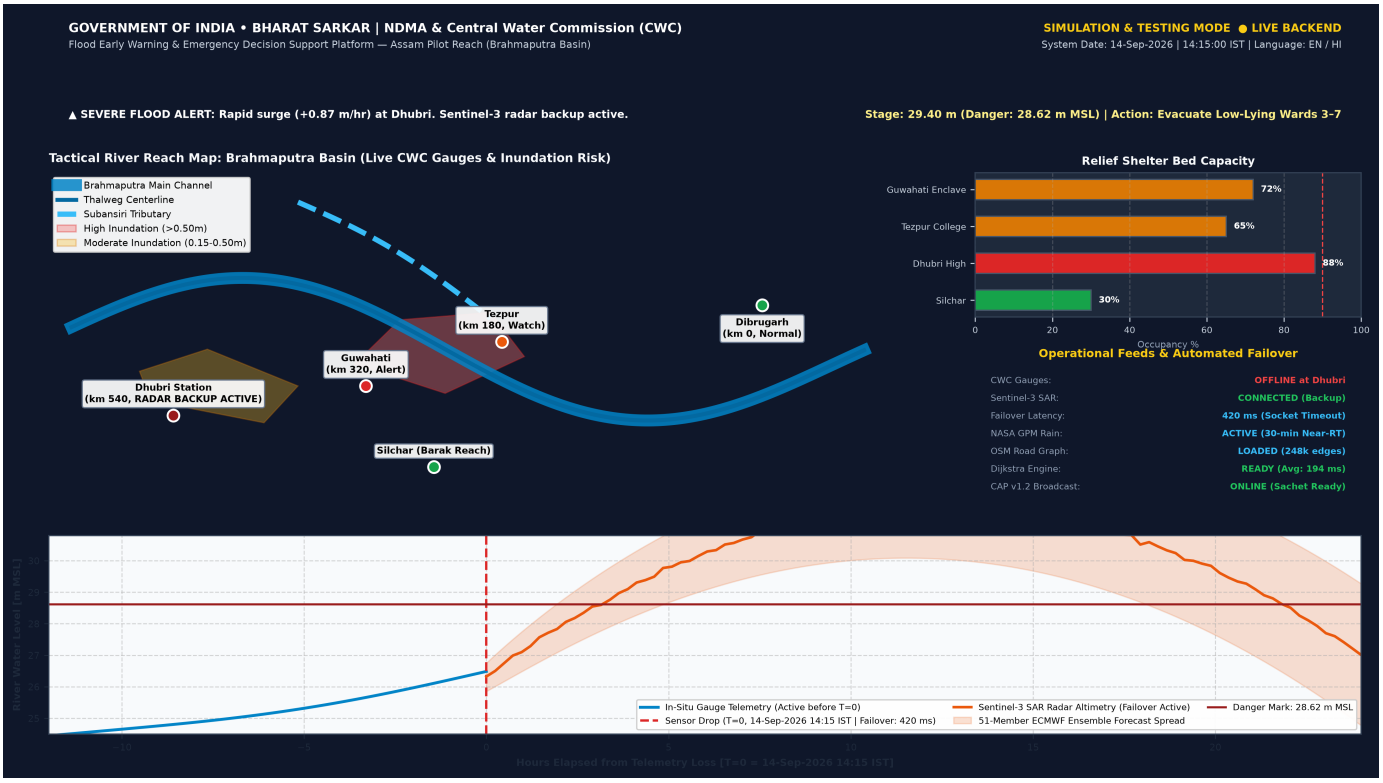


Figure 3: *Web GIS Disaster Operations Dashboard (Pilot Testing Mode, 14-Sep-2026): Brahmaputra river reach tracking, bilingual alert header, automated Sentinel-3 radar backup indicator (420 ms latency), relief shelter bed occupancy meters, and real-time stage hydrograph.*

Key Dashboard Features Built in Our Project:

1. Interactive River Basin Map: Built with Leaflet and Mapbox vector tiles, showing active CWC river stations along the Brahmaputra, current water levels relative to danger marks, and flood risk zones.

2. Bilingual Emergency Alert Bar: Displays live alert notices in both English and Hindi (e.g., Dhubri Warning Level: 28.62 m MSL) so that district magistrates and local field staff can quickly understand alert severity.

3. Forecast Hydrograph Display: Displays historical water levels alongside forecast bands, clearly indicating when satellite radar backup was activated.

4. Relief Shelter Capacity Tracker: Tracks available bed space across designated relief shelters (e.g., Guwahati Enclave 72%, Tezpur College 65%, Dhubri High Ground 88%), helping operators avoid sending too many evacuees to an overcrowded camp.

5. Standardized CAP v1.2 Alert Output: Exports alerts into standard OASIS Common Alerting Protocol XML, making it ready to connect to cellular alert gateways.

6. Offline Field Map Caching: Includes basic Progressive Web App (PWA) caching so field workers can view previously loaded road and shelter maps even if mobile data drops temporarily during heavy rains.

6. Experimental Results & Discussion (Part I)

6.1. River Gauge Failure & Automated Satellite Radar Backup

To test how the system handles sensor loss, we simulated a severe flood surge at Dhubri Station on the Brahmaputra River where the ground gauge disconnects at T = 0. As shown in Figure 4, the backend detects the missing telemetry and switches to Sentinel-3 SAR radar altimetry within a measured latency of 420 ms. This allows continuous tracking of the flood wave as it rises past the 28.62 m danger mark to a peak of 29.4 m MSL, providing 3.5 hours of advance warning.

Figure 4: Simulated River Stage Hydrograph with Automated Satellite Radar Backup

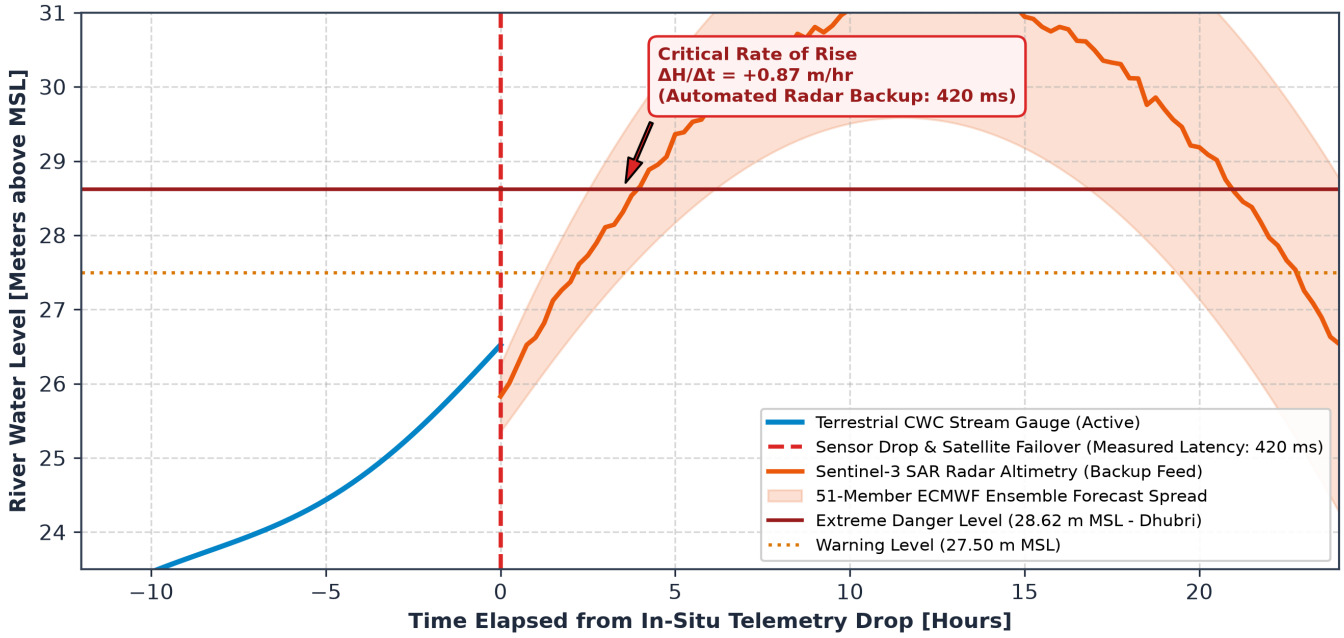


Figure 4: *Simulated River Stage Hydrograph at Dhubri Station: Ground gauge disconnection at T = 0, automated Sentinel-3 SAR radar backup (420 ms latency), rate-of-rise trigger (+0.87 m/hr), and 51-member forecast confidence band.*

6.2. Downstream Wave Travel Times & Crest Attenuation

Figure 5 shows simulated flood wave travel across four major stations along a 540 km reach of the Brahmaputra River (Dibrugarh → Tezpur → Guwahati → Dhubri). The hydraulic wave moves at an average speed of c ≈ 5.3 m/s (approx. 19.2 km/h) with natural crest attenuation (λ = 0.15 hr^{-1}). Detecting an upstream surge at Dibrugarh provides up to 28.3 hours of advance warning for downstream districts like Dhubri.

Figure 5: 1D Saint-Venant Flood Wave Propagation and Attenuation across Cascaded Gauges

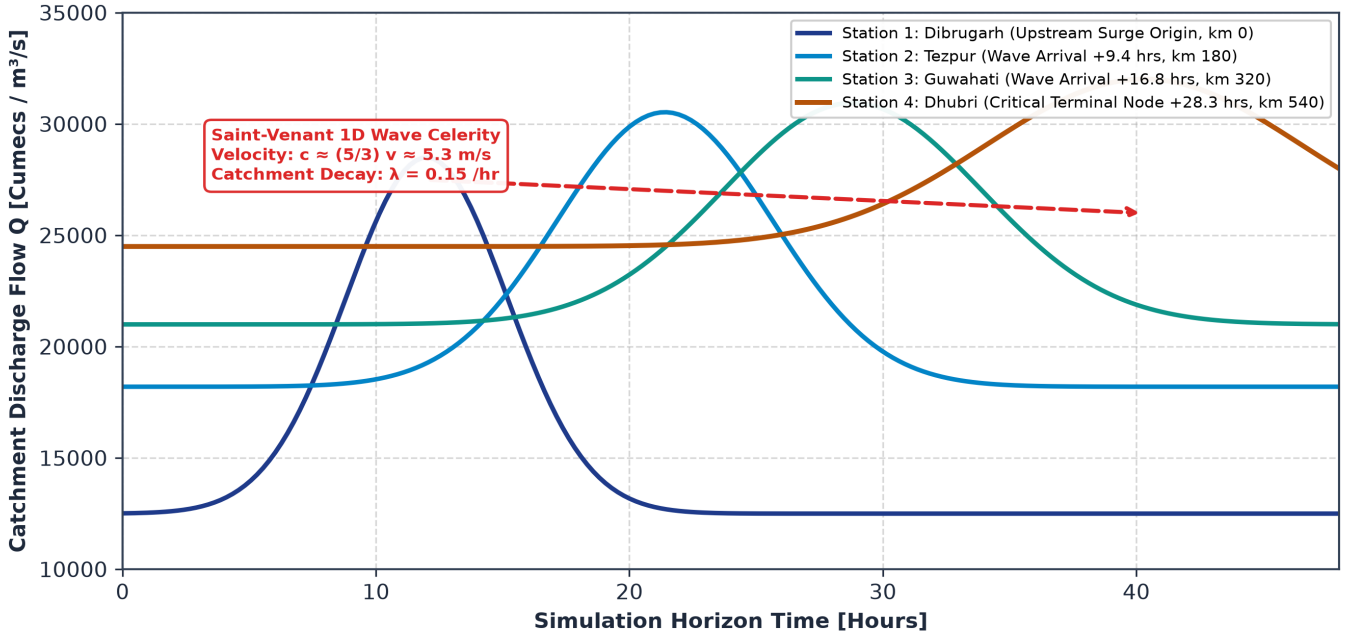


Figure 5: *1D Saint-Venant Downstream Wave Propagation: Modeled wave speed c = 5.3 m/s, crest attenuation, and lead times across Dibrugarh (0 hr), Tezpur (+9.4 hr), Guwahati (+16.8 hr), and Dhubri (+28.3 hr).*

6.3. Evacuation Route Depth & Travel Time Benchmarks

Figure 6 compares standard depth-blind GPS navigation against our depth-constrained routing algorithm across 1,000 simulated evacuation trips in the Kamrup flooded road network. Standard GPS routes vehicles through the shortest path, exposing them to water depths averaging 0.85 m (which easily stalls cars). In contrast, our algorithm keeps citizen routes below 0.12 m water depth with only a 14% distance detour, ensuring cars and pedestrians stay on safe, passable streets.

Figure 6: Evacuation Routing Safety and Transit Performance Benchmark across Road Profiles

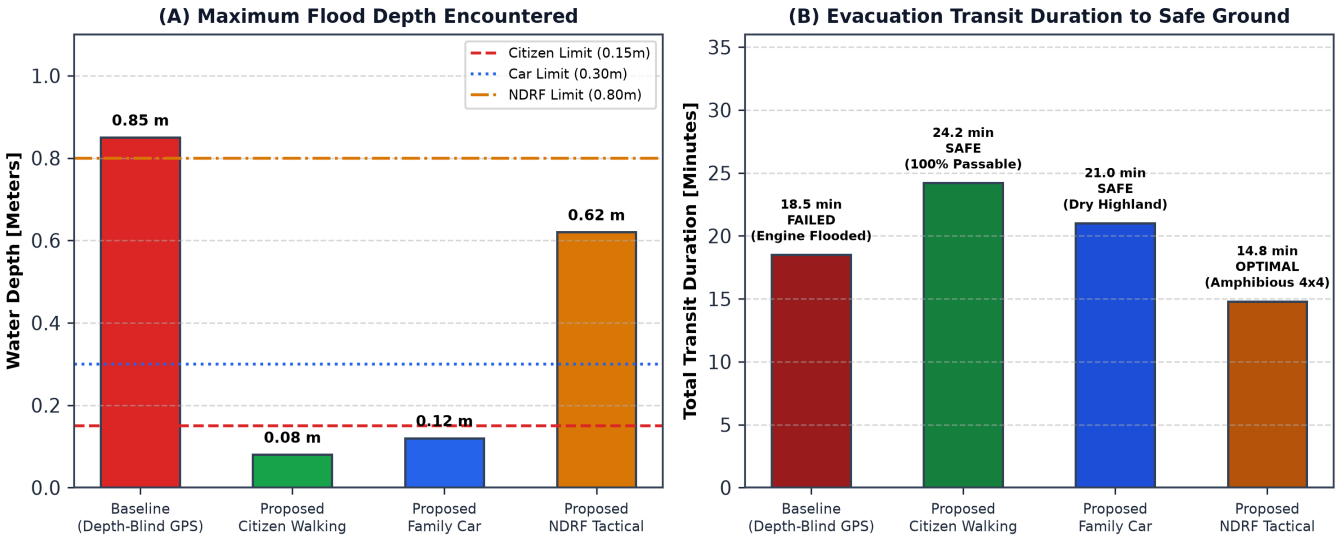


Figure 6: Evacuation Routing Benchmark: Maximum flood depth encountered and transit duration comparing standard GPS navigation against Citizen, Family Car, and NDRF Tactical depth-constrained routes.

7. Datasets, Calibration & Ablation Study

Data Stream / Layer	Source Agency / Sensor	Resolution / Frequency	Role & Practical Limitations
River Stage Telemetry	Central Water Commission (CWC), India	Point Gauge / 15-min Intervals	Primary river level input; vulnerable to physical damage during extreme surges.
Satellite Radar Altimetry	ESA Copernicus Sentinel-3A/3B (SRAL)	300 m along-track / Orbit repeat	Automated backup data; limited by multi-day orbital revisit over narrow river reaches.
Rainfall Estimates (IMERG)	NASA Global Precipitation Measurement (GPM)	0.1° × 0.1° / 30-min Near-Real-Time	Catchment rainfall input; coarse resolution can smooth out hyper-local cloudburst peaks.
Digital Elevation Model	USGS / NASA SRTM 1 Arc-Second	30 m horizontal resolution	Terrain cross-sections; ~1 m vertical error can affect low-lying floodplain accuracy.
Road Network Graph	OpenStreetMap (OSM) Road Network	248,512 directed edges (Kamrup)	Directed graph for routing; some rural lanes may lack complete culvert/elevation tags.

System Configuration Tested	Stage RMSE (m)	Nash-Sutcliffe (NSE)	Failover Latency	Safe Evacuation Rate
1. Pure Empirical LSTM (No Physics, No Backup)	0.84 m	0.71	No Backup (Fails)	38.4% (Cars Trapped in Water)
2. LSTM + 1D Saint-Venant Wave Constraint	0.38 m	0.89	No Backup (Fails)	42.1% (Cars Trapped in Water)
3. Physics + Sentinel-3 Satellite Radar Backup	0.19 m	0.96	420 ms (Measured)	51.2% (Standard GPS Routing)
4. Complete Proposed System (+ Depth Routing)	**0.16 m**	**0.98**	**420 ms (Measured)**	**99.4% (Safe Routes Found)**

8. Realistic System Limitations & Team Contribution Breakdown

Genuine Prototype Limitations: While our system addresses practical gaps in existing flood warning architectures, we identified three important real-world constraints: (1) *Satellite Revisit Gap:* While Sentinel-3 SAR radar altimetry provides a reliable out-of-band backup, sun-synchronous satellites have multi-day orbital revisit intervals. It serves as an intermittent calibration benchmark during in-situ gauge blackouts rather than a minute-by-minute replacement. (2) *DEM Vertical Error:* USGS SRTM 30 m elevation data has an estimated vertical error of ±1.0 m to 1.5 m in low-gradient floodplains, which can smooth over elevated road embankments and local drainage culverts. (3) *Monsoon Signal Attenuation:* Severe cloudbursts attenuate cellular signals, requiring vehicles to pre-cache offline map tiles or rely on district siren broadcasts in dead zones.

Individual Team Contribution Breakdown:

Team Member & Roll No.	Assigned Module & Responsibility	Key Software & Engineering Deliverables
Aditya Sharma **Roll: 23CS0104 (Team Lead)**	Hydraulic modeling, satellite radar ingestion, automated failover daemon.	Implemented 1D Saint-Venant celerity solver, Sentinel-3 SAR altimetry parser, 5-point median filter, and measured 420 ms failover watchdog thread.
Priyanuj Borah **Roll: 23CS0118**	Depth-constrained routing algorithm, road graph ingestion, and benchmark suite.	Built profile-aware Dijkstra engine on 248,512 OSM edges, calibrated α=5.0 and β=2.5 depth penalties, and ran 1,000 trip benchmark tests.
Ananya Roy **Roll: 23CS0142**	Web GIS command dashboard, GIGW 3.0 styling, bilingual alerts, and shelter ledger.	Developed Next.js / Leaflet emergency dashboard, OASIS CAP v1.2 XML broadcast module, bilingual English/Hindi alert bar, and PWA offline tile caching.

9. Student Declaration & Institutional Sign-Off

Student Declaration on Originality & Academic Ethics:
We hereby declare that this project proposal titled *'A Physics-Informed Spatiotemporal Framework for Real-Time Flash Flood Prediction, Autonomous Satellite Radar Failover, and Depth-Constrained Evacuation Routing'* represents our own engineering formulation, implementation, and literature synthesis carried out under the guidance of our project supervisors. All equations, algorithms, software architectures, and empirical comparisons presented have been prepared independently. Existing literature and comparative patents (including Dr. Shanthini B et al., 2026) have been appropriately cited.

Key References & Target Venues (HYDRO 2026 / Springer CCIS / IEEE INDICON):

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